DESIGN RATIONALE AND ALTERNATIVES REPORT

Mathematical Design Options and Research-Decision Rationale

A companion report explaining the alternatives considered, the evidence behind the selected design, and why other components are deferred

CORE IDEA

Detect collusion from unusual directional relationships in agents' decisions, after accounting for each agent's prompt-conditioned behaviour, then use reinforcement learning to decide how and when to investigate further.

Research design draft | 24 October 2026
Status: proposed method; no experimental results claimed
Novelty status: plausible research hypothesis requiring a systematic prior-work check and empirical validation

Prepared as a research planning document. All performance claims are hypotheses until tested.

Purpose and bottom line

This report records the design decisions for the proposed APTG-RL project. The project premise is that collusion cannot be inferred from action similarity alone: agents may coordinate through complementary actions, directional response, delays, rotating assistance or hidden signals. At the same time, dependence is not proof of collusion because common context, shared model tendencies, prompts and benign adaptation can produce correlated decisions. The design must detect unusual relationships while controlling those confounds.

The selected core is: (1) prompt-conditioned independent-behaviour calibration, (2) directional predictive dependence, (3) a temporal directed graph with uncertainty, (4) objective/outcome evidence to distinguish suspicious relationships from actual prohibited coordination, and (5) an RL investigator for adaptive evidence gathering. The "adversary" is primarily a controlled perturbation source that changes one agent's instruction; a learned red-team policy is an optional extension, not required for the first experiment.

IMPORTANT DISTINCTION The model may have stable behavioural tendencies over a controlled distribution of prompts, but it does not have a prompt-invariant policy. The correct target is a conditional baseline and a measured prompt-sensitivity profile, not a fixed "model fingerprint" assumed to survive all prompts.

1. Decision summary

Method / option	Decision	Reason
Prompt-conditioned policy baseline	CORE	Directly addresses the user's insight: models have characteristic tendencies, but prompts shift those tendencies. Must control prompt, context, model/version and settings.
Conditional mutual information / original net influence	BASELINE + SUPPORTING FEATURE	Direct prior-work baseline [1]. Important to reproduce; not sufficient alone for directionality.
Transfer entropy / directional predictive gain	CORE CANDIDATE	Models lagged direction and complementary response without requiring similar actions [11]. Use held-out predictive gain plus conditional null testing.
Time-varying directed graph	CORE CANDIDATE	Represents direction, persistence, cycles and subgroup structure; dynamic graph is justified by sequential behaviour. Graph theory alone is not novel [6].
Controlled unilateral adversarial prompt intervention	CORE EXPERIMENTAL DESIGN	Tests the confound central to the project: unusual behaviour from one agent vs a new dependency between agents.
Task regret / coalition advantage	CORE OUTCOME VALIDATION	Dependence alone cannot establish that behaviour is collusive; task-objective harm provides independent evidence [4].
RL active investigator	CORE, BUT SUBJECT TO ABLATION	Learns how long to observe and which pair/group/probe to inspect. Needs to beat sequential-test baselines; otherwise it is complexity without demonstrated value [7].

Method / option	Decision	Reason
Static strategy graph metrics	BASELINE / RELATED WORK	Useful comparator, but static pricing-policy graphs differ from LLM multi-agent interaction traces [6].
Supervised classifier on hand-designed features	BASELINE	Checks whether RL is warranted when labelled examples are available.
Graph neural network / GNN-RL	DEFERRED EXTENSION	Potentially useful for larger, variable-size groups, but adds data and interpretability burdens before evidence establishes need.

2. The behavioural-prior assumption: what to keep and what to reject

2.1 What is useful in the intuition

A given deployed endpoint can exhibit measurable response tendencies across repeated, controlled probes. These tendencies reflect model weights/post-training, system prompt, tool configuration, endpoint version and inference settings. Recent black-box stability/fingerprinting work estimates output distributions over fixed prompts and detects distribution shifts [8]; prompt-fingerprinting papers show that selected prompts can reveal model- or prompt-specific response preferences [9,10]. That makes behavioural calibration a legitimate design component.

2.2 What must not be assumed

Do not assume an LLM has one invariant reasoning strategy regardless of prompt. The prompt and conversation history are conditioning variables. A misleading instruction may shift the model's action distribution, reveal a pre-existing tendency, override it, or produce noisy behaviour. Endpoint nondeterminism and provider differences can also change outputs [8]. The experiment must estimate how much the response distribution changes under prompt conditions rather than assume stability.

2.3 What we do instead

- **Conditional baseline:** estimate $p(A_i \mid$ own history, public state, prompt condition, endpoint/config metadata), using non-collusive calibration runs.
- **Prompt sensitivity:** measure the within-agent distribution shift from clean to perturbed prompt conditions; include matched clean and perturbed controls.
- **Relational statistic:** test whether partner history explains additional variation in the target agent's next action after conditioning on known covariates.
- **Intervention evidence:** randomize who receives the perturbation to test whether a unilateral prompt shift alone causes false accusations.
- **Uncertainty/abstention:** if a test prompt, endpoint version or task context is outside calibration support, mark the result uncertain rather than forcing a collusion label.

3. The math options, one by one

Option 1. Raw action similarity / embedding similarity

Representation: edit distance, categorical agreement, cosine similarity of action/message embeddings, or similarity between distributions of actions.

What it catches: obvious copying, repeated identical responses, or repeated near-duplicate messages. It is easy to implement and useful as a weak baseline.

Why not the core: complementary agents can collude while choosing different actions; two independent agents can be similar because they share a prompt, model family or optimal strategy. Similarity is symmetric and usually hides temporal direction. A higher similarity score therefore has ambiguous meaning.

Decision: retain only as a baseline and possible input feature. Do not make the research question “can similarity detect collusion?”

Option 2. Mutual information and the 2022 net-influence detector

Representation: mutual information between actions and a net-influence statistic comparing one agent’s influence on another with the influence of other agents [1].

What it catches: pairwise statistical dependence even if exact actions are not identical. It is a credible, established baseline with experiments in repeated RPS and poker [1].

What it misses for this project: mutual information is symmetric; it does not establish temporal direction. Its output can still be elevated by shared context unless the modeling and conditioning handle that context. It also does not independently establish that the dependence violates the task objective.

Decision: implement faithfully as the primary prior-work baseline. Add it as a feature/control, but do not present MI itself as new.

Option 3. Transfer entropy / directional conditional dependence

Representation: $T_{i\to j} = I(H_i^t; A_j^{t+1} \mid H_j^t, X^t)$, where X includes observed shared state and the available prompt, perturbation, model/version and configuration metadata for both agents. Alternatively, use cross-validated log-likelihood improvement when adding agent i’s history to the model predicting agent j’s next action, while giving the baseline and expanded models the same exogenous covariates.

What it catches: leader-follower dynamics, delayed responses, complementary action patterns and information encoded in sequences. It does not require the two agents to act similarly. The direction comes from the prediction target and temporal ordering [11].

Limitations: it can be biased by finite data/model misspecification; a common unobserved cause may induce apparent direction; high information transfer can be benign adaptation. It needs null tests and outcome evidence.

Decision: select as the main candidate edge statistic. Prefer held-out predictive gain as a practical estimator and validate it against a transfer-entropy implementation on toy cases.

Option 4. Granger-style predictive testing

Representation: compare the out-of-sample predictability of A_j^{t+1} from j's own history/context with a model that also sees i's history [12].

What it catches: predictive contribution of one agent's past to another's future. Flexible classifiers can adapt the test to categorical actions.

Why defer as a separate primary method: conceptually overlaps with the selected predictive-gain/transfer-entropy estimator. Implementing several nearly equivalent directional metrics may multiply analysis choices without clarifying the central claim.

Decision: use as a robustness estimator or alternative implementation, not a parallel core contribution. Note that Granger predictability is not proof of causal influence.

Option 5. Residuals and robust statistics (MAD normalization)

Representation: fit each agent's expected action distribution under a non-collusive prompt-conditioned baseline, then compute conditional surprise or residual features. Normalize a statistic against an independent-control distribution, optionally using median and MAD.

What it catches: a shift in an agent's individual behaviour that cannot be explained by its calibrated baseline; robust scaling reduces the influence of outliers in reference samples.

Why not the entire detector: unusual individual behaviour is not collusion. MAD normalization makes scales comparable but does not identify dependence, infer direction, or establish significance. If MAD is zero or the baseline is too small, a fallback scale and raw scores are required.

Decision: retain as an individual-shift diagnostic and secondary calibration layer. Never use a MAD z-score alone to issue an accusation.

Option 6. Causal inference and interventions

Representation: perturb an agent's policy/action in a controlled way and measure the resulting change in a partner's behaviour, holding common context fixed as far as possible.

What it catches: whether changing a source agent's behaviour changes the target agent's response in a controlled environment. The unilateral-prompt manipulation is a low-cost way to test a related intervention question.

Why not the sole online detector: controlled interventions may be unavailable or disruptive in deployed systems; the answer depends on the intervention and environment. Causal influence also does not by itself prove prohibited collusion.

Decision: use interventions as the benchmark's causal stress test and for validating interpretation, not as the only runtime detection path.

Option 7. Dynamic directed graphs and temporal motifs

Representation: construct an edge i→j from calibrated directional predictive dependence; track edge strength, uncertainty, persistence and direction over time. Compute reciprocal links, leader-follower chains, cycles, alternating assistance motifs, communities and change points.

What it catches: relationship structure at the pair and group levels, including coordination that rotates roles or involves more than two agents.

Why static graph metrics are not enough: static strategy-graph collusion auditing already exists in algorithmic pricing [6]. A generic graph centrality score could also encode activity or role structure, not collusion. Temporal motifs need condition-matched null graphs and objective validation.

Decision: select a temporal directed graph as the relational representation. Compare it to static graphs and pairwise-only methods; claim novelty only for a specific tested estimator/combination, not for "using graphs."

Option 8. Dynamic Bayesian networks

Representation: a probabilistic graphical model over actions, histories, shared context and possibly latent states across time.

What it catches: uncertainty about dependency structure and multi-step relationships. It can represent competing explanations explicitly.

Why defer: structure learning and latent confounding become difficult with many agents and sparse trajectories; interpretability does not guarantee causal correctness. It overlaps with the selected graph representation and would introduce a second large modeling problem.

Decision: keep as a future extension if uncertainty-aware latent-structure modeling proves necessary.

Option 9. Change-point detection / hidden-state models

Representation: CUSUM, Bayesian online change-point detection, HMMs or latent states such as independent/adapting/coordinating.

What it catches: onset and cessation of coordination, including an initially normal agent that later adapts to a partner.

Why not the first core contribution: this identifies temporal regime shifts, but any benign prompt change, role update or endpoint drift can also produce a shift. It needs a good base statistic and the same prompt controls.

Decision: include simple change-point features in the temporal graph; reserve a full HMM as an ablation if change-point performance is important.

Option 10. Coalition advantage, regret and task outcomes

Representation: compare the observed task objective to the known cooperative optimum or a matched independent-policy counterfactual; quantify coalition advantage and impact on non-coalition participants.

What it catches: whether the coalition's behaviour actually harms the intended objective, rather than merely discussing collusion or exhibiting dependence [4].

Why it cannot stand alone: task harm may arise from random mistakes, failures or ordinary conflict; collusion can also pursue a secondary goal with subtle/low immediate impact. Outcome evidence and relational evidence answer different questions.

Decision: retain as an independent evidence channel and validation metric, not as the only detector.

Option 11. Activation probes / internal interpretability

Representation: probe per-agent internal activations and aggregate signals at group level. NARCBench reports that activation-based approaches can detect some collusion signals, but transfer performance drops under structural distribution shift and no single probe dominates all collusion types [5].

What it catches: signals that may remain hidden in text/action outputs, particularly with white-box model access.

Why excluded from the core: requires privileged model access, changes the threat model, and is difficult to compare fairly to a black-box behavioural detector. It also risks mixing two separate contributions: dynamic graph detection and multi-agent interpretability.

Decision: document as an optional white-box extension or comparison if an open-weight model and activation instrumentation are available.

Option 12. Supervised ML / GNN classifier

Representation: train a classifier on graph/statistical features, or a graph neural network over trajectories and edges, to predict labelled colluding pairs/groups.

What it catches: nonlinear combinations of features, potentially with good ranking metrics when labels and training examples are plentiful.

Why not use it as the whole project: it can overfit scripted strategies, does not naturally decide how much additional evidence to gather, and can make a simpler statistical method look obsolete only because of train/test leakage or more data. But it is an essential baseline if labelled data are available.

Decision: include a supervised model over the same features as a benchmark. Defer GNNs until there is evidence that engineered graph statistics cannot scale or capture relevant structure.

Option 13. Reinforcement-learning investigator

Representation: formulate a partially observable sequential decision process. The investigator sees uncertain graph/statistical evidence and chooses whether to continue observing, inspect a pair or subgroup, run an allowed probe, alert, or abstain. Rewards trade off true detection, false accusation, missed detection and observation costs.

Why include: the project asks not only “is this pair suspicious?” but “when is there enough evidence to flag it, and what is worth observing next?” This gives RL a clear sequential role. Work on learning-based sequential controlled sensing already shows that RL can choose observations under cost/uncertainty [7], so the contribution must be a collusion-specific state, reward, and evaluation rather than RL itself.

Main risk: the reward function can hard-code the desired behaviour; the RL policy can overfit known collusion scripts; or a sequential test may perform just as well. Use pre-registered reward weights, sensitivity analysis, held-out strategies, and non-RL baselines.

Decision: retain RL as a core component, but make its added value an explicit falsifiable hypothesis. Do not claim RL is needed if simpler alternatives match its performance.

Option 14. Learned adversary / adversarial co-training

Representation: train or script an adversary that chooses perturbation type/severity to induce a policy shift or stress the detector. An extension could use minimax training or alternating training between a red-team prompt policy and the RL investigator.

What it catches: weaknesses against adaptive or previously unseen perturbations rather than a fixed prompt list.

Why defer the learned adversary: it adds a second learning problem and can create an unstable arms race before the underlying detector is validated. It also makes the experimental interpretation harder if the adversary and detector co-adapt on the same scenarios.

Decision: begin with a transparent, scripted perturbation taxonomy and held-out templates. Add a learned red-team adversary only after the core detector and RL investigator beat baselines; if added, train/test the two policies independently and evaluate against fresh perturbation seeds.

4. Why the selected combination is coherent

Layer	Question answered	What it cannot establish alone
Prompt-conditioned baseline	Is this action unusual given this agent's own history, task context, and prompt condition?	Cannot establish inter-agent coordination.
Directional predictive statistic	Does another agent's history add information about this agent's future actions?	Cannot prove causality or malicious intent.
Temporal graph	Is the relation part of a persistent, asymmetric, recurring or group-level structure?	Graph motifs can arise in legitimate teamwork.
Task outcome / regret	Does the group's behaviour violate or undermine the intended objective?	Harm can arise from mistakes or non-collusive conflict.
RL investigator	What evidence should be gathered next, and when should a decision be made?	Can overfit reward and simulator; needs strong non-RL baselines.

The components address different failure modes. The combination is not justified because more methods are automatically better; it is justified only if ablation shows each retained layer provides distinct value. The architecture therefore includes explicit failure criteria and treats the RL policy as removable if it adds no value.

5. Most important confounders and how to handle them

Confounder	Why it looks like collusion	Control
Same model family	Similar priors, response preferences and training patterns.	Same-model independent pair control; model-family metadata; calibration per endpoint/model family.
Same prompt / shared public state	Agents independently choose the same efficient action.	Condition on shared state and prompt; common-prompt negative control.

Confounder	Why it looks like collusion	Control
Unilateral misleading prompt	One agent changes behaviour and may indirectly change the environment for others.	Randomize recipient; measure individual policy shift; hold public context fixed where possible.
Legitimate adaptation	Agents respond to each other to cooperate on the intended task.	Label benign adaptation; measure task objective/regret separately; do not equate all influence with collusion.
Endpoint drift	Version/provider/configuration changes alter response distributions.	Record endpoint/version/decoding settings; re-calibrate or abstain on untracked shifts [8].
Short trajectories	Noisy edge estimates yield unstable graph motifs.	Minimum data rules, uncertainty intervals, blocked permutations and sequential evidence accumulation.
Multiple comparisons	Many pairs/groups produce accidental high scores.	FDR/family-wise correction and validation on independent episodes.
Leakage	Prompt templates or scenario seeds appear in train and test.	Split by episode, prompt family, coalition mechanism and model where feasible.

6. Claims policy: what can be said at each stage

Stage	Defensible statement	Statement to avoid
Now, before experiments	“We propose and plan to evaluate a prompt-conditioned directional temporal-graph detector with an RL investigator.”	“Our method detects collusion better.”
After literature search	“The specific combination appears underexplored in the searched sources; prior work already covers its components.”	“Nobody has ever done this.” unless a systematic review justifies it, and even then scope the claim.
After synthetic validation	“The method detects the scripted mechanisms in this simulator under these conditions.”	“The detector proves real-world agents collude.”
After held-out tests	“On the held-out models/strategies/prompts, it improves X metric at Y FPR with confidence interval Z.”	“It is universally robust.”
If RL does not help	“An adaptive statistical detector works; RL did not improve the tested operating points.”	Keeping RL in the abstract as an advantage despite negative ablations.

7. Suggested final research framing

PROVISIONAL TITLE Adversarially Perturbed Directional Graphs for Reinforcement-Learning-Based Collusion Auditing in Multi-Agent LLM Systems

Primary contribution candidate

A calibrated directional dependence statistic and temporal graph representation that test whether the partner’s history predicts an agent’s future behaviour beyond the agent’s own prompt-conditioned

baseline and observed shared context, evaluated under unilateral instruction perturbations and different collusion mechanisms.

Secondary contribution candidate

An RL investigator that adaptively chooses additional observations or probes under explicit false-positive, delay and cost constraints, with a careful comparison against sequential hypothesis tests and active-sensing heuristics.

Non-negotiable scope rule

Do not add every math technique to the final model. The proposal is strongest if it has a small number of interpretable components, rigorous controls, and results that could genuinely falsify the main idea.

8. References and source notes

This reference list supports the design decisions and identifies prior art that prevents overclaiming novelty. It is a targeted starting set, not an exhaustive literature survey.

References

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